

Murari Ganesan

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Education

University of California, Berkeley, BS in Data Science and Economics – Berkeley, CA Jan 2023 – May 2026

• Relevant coursework: Machine Learning, Deep Learning, Causal Inference, Econometrics, Probability & Statistics

Research Experience

Research Assistant, Berkeley AI Research (BAIR) — Serina Chang Lab – Berkeley, CA Aug 2025 – present

Investigating whether large language models can predict public opinion on novel real-world events, with a focus on computational social science applications. Work connects to questions of how information spreads and is processed within social systems.

- Designed experimental pipeline scraping temporally accurate Wikipedia event contexts and eliciting model outputs across multiple formats (next-token probabilities, verbalized JSON, free-text)
- Evaluated LLM opinion prediction using survey benchmarks, measuring error, Wasserstein distance, and KL divergence across events and model families
- Analyzed model stability, systematic biases, and failure modes to support reproducible CSS research

Research Assistant, University of California, Berkeley — Department of Economics – Berkeley, CA June 2024 – June 2025

Supported empirical economics research spanning political corruption and female labor outcomes, using quasi-experimental and experimental methods.

- Conducted regression analysis and difference-in-differences estimation on political corruption data using Python, pandas, and R
- Built automated data pipelines using Selenium and Pytesseract to collect and parse legislative records
- Assisted with research design and analysis for a randomized controlled trial studying female labor market outcomes in India
- Co-authored sections of research papers summarizing empirical findings

Industry Experience

Forward Deployed Engineer, Snorkel AI – Redwood City, CA June 2025 – present

Developed LLM-based evaluation infrastructure and data pipelines for enterprise machine learning workflows.

- Built and deployed LLM-as-a-judge evaluation systems for expert contributor quality assessment
- Designed internal benchmarking suite using novel questions to evaluate frontier LLM model performance
- Implemented data engineering pipelines to transform and format training data for client model fine-tuning

Projects/ Papers

Beyond the Cutoff: Evaluating LLMs for Public Opinion Prediction on Unseen Events 2025–2026
(Data Science Honors Thesis)

Evaluated how large language models estimate public opinion on newly emerging events — directly relevant to understanding how information and beliefs propagate in networked social systems.

- Built end-to-end pipeline connecting real-time event data (Wikipedia revisions), LLM inference, and survey benchmarks
- Compared multiple elicitation formats and model families; documented systematic biases and calibration failures
- Applied information-theoretic metrics (KL divergence, Wasserstein distance) to assess distributional alignment with human survey responses

Trade Disruption Following Francis Scott Key Bridge Collapse

Spring 2025

Causal analysis of a sudden infrastructure shock on regional trade network flows in the Chesapeake Bay using difference-in-differences.

- Identified causal effect of bridge collapse on cargo volumes and route reallocation across the eastern seaboard
- Applied parallel trends diagnostics and robustness checks to strengthen identification strategy

Customer Service Request Classification via NLP

Summer 2025

End-to-end NLP pipeline combining LLM-based weak labeling with neural network transfer learning to classify high-volume service requests.

- Used a local LLM to generate initial labels on a subset of data, then trained a fine-tuned BERT-based classifier for weekly/monthly inference
- Reduced manual labeling burden by over 80% while maintaining high classification accuracy

Skills

Econometrics & Methods: Difference-in-differences, regression analysis, RCT design, causal inference, survey data analysis

Programming: Python, R, SQL, C++, Java, Git

ML & Data Science: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, pandas, NumPy, SciPy, matplotlib

Data Collection: Selenium, Pytesseract, web scraping, API integration, large-scale dataset construction